

Assignment:-2

Task 1:- List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

No.	Common Hardware Issue	Typical Symptom
1.	RAM failure	PC may not boot, show a blank screen, or produce repeated beep sounds.
2.	Hard disk/SSD failure	System becomes very slow, files may not open, or the PC shows a “No boot device” error.
3.	Power Supply Unit (PSU) failure	PC does not turn on, shuts down suddenly, or restarts unexpectedly.
4.	CPU overheating	PC becomes very hot, slows down, freezes, or shuts down automatically.
5.	Graphics card failure	No display, screen flickering, graphical artifacts, or system crashes during graphics-intensive tasks.

Task 2:- Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

- Check the power socket

Plug another device into the socket.

Check: Make sure the socket is working.

- Check the charger

Look at the charger and cable.

Check: Look for cuts, broken wires, or damage.

- Check the charging port

Look inside the laptop charging port.

Check: Make sure there is no dust or damage.

- Connect the charger properly

Plug the charger firmly into the laptop and socket.

Check: See if the charging light turns on.

- Try another charger

Use another compatible charger if available.

Check: If the laptop charges, the old charger may be faulty.

- Check the battery

If the battery can be removed, remove it and connect the charger.

Check: If the laptop turns on, the battery may have a problem.

- Check for overheating

Touch the charger carefully and check for unusual heat or burning smell.

Check: If there is too much heat or a burning smell, stop using the charger.

- Use a technician if needed

If the laptop still does not charge, the charging port, battery, or internal hardware may be faulty.

Task 3:- Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Testing USB Charger Voltage with a Multimeter

1. Take a USB phone charger and plug it into the power socket.
2. Set the multimeter to DC Voltage (V=).
3. Set the range to 20V DC if the multimeter is not automatic.
4. Put the black probe on the USB GND (-) pin.
5. Put the red probe on the USB 5V (+) pin.
6. Read the voltage shown on the multimeter.

Measured voltage: 5.02V

Expected range: 4.75V to 5.25V

Result: 5.02V is within the expected range, so the charger voltage is normal.

Task 4:- A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Testing the PSU 12V Rail – Simple Steps

1. Turn off the PC and unplug it from the power socket.
2. Disconnect the ATX connector from the motherboard.
3. Set the multimeter to DC Voltage (V_{DC}).
4. Connect the black probe to a black wire (GND).
5. Connect the red probe to the yellow wire (+12V).
6. Read the voltage on the multimeter.

Measured voltage: 12.08V

Expected voltage: Around 12V

Result: 12.08V is normal.